

A dynamic graph-based PU-learning approach for sequential recommendation in rare disease detection

Guanhao Wei^{1,*}, Yunlong Wang¹, Li Zhou¹

Academic Editor: Victor Sheng

Abstract

Introduction: Rare diseases affect over 300 million people globally; their low prevalence often leads to high misdiagnosis rates. While Electronic Health Record (EHR) data offers opportunities for early detection, existing predictive models struggle to capture the complex, time-sensitive sequence of patient journeys.

Materials and methods: To address this, we propose a Dynamic Graph-based Positive-Unlabeled Learning approach (DPU-SR) to forecast patient trajectories and identify undocumented rare diseases. Our framework reconstructs patients and clinical events into dynamic subgraphs, utilizing time-aware information propagation to evaluate how sequential clinical actions influence health outcomes. To address EHR limitations where records indicate diagnosed conditions but cannot definitively rule out others, we implement dynamic structure-aware negative sampling within a Positive-Unlabeled (PU) learning setting.

Results: After benchmark validation, we deployed the DPU-SR algorithm on a real-world EHR dataset of approximately 1.5 million candidates to detect unrecognized multiple myeloma (MM). By capturing the temporal evolution of clinical interactions, our approach achieved an average 6.3% precision among the top 1000 scored patients from each testing cohort, showing a 20% improvement over the commonly used production models in the healthcare industry.

Conclusions: Transforming sequential EHR data into dynamic graphs provides an effective, scalable artificial intelligence tool for population-wide disease surveillance. Enhancing early detection precision provides significant predictive enrichment and the potential to inform early intervention, offering a scalable foundation to support public health practice, optimize resource allocation, and proactively manage patient care.

Keywords: *electronic health records, graph neural network, disease detection, artificial intelligence, positive unlabeled learning*

Citation: Wei G, Wang Y, Zhou L. A dynamic graph-based PU-learning approach for sequential recommendation in rare disease detection. *Academia AI and Applications* 2026;2. <https://doi.org/10.20935/AcadAI8326>

1. Introduction

It has been estimated that more than 300 million people worldwide are living with one of the approximately 7000 rare diseases. Due to their low prevalence, it is hard for healthcare providers to easily find a common significant representation of specific diseases, resulting in a high misdiagnosis rate. The earlier patients can receive an accurate diagnosis, the higher the survival rate they will have, and fewer medical resources will be wasted. The availability of mass Electronic Health Records (EHRs) provides the possibility for researchers to leverage modern machine learning algorithms, such as Recurrent Neural Networks (RNN), Generative Adversarial Networks (GAN), and Convolutional Neural Networks (CNN), to make precise predictions for disease detection tasks. Because a patient's medical journey contains complex, longitudinal information, including demographics, disease diagnosis, medical treatment procedure, and prescription drug data, modeling these trajectories requires highly sophisticated sequence processing. Recent advancements in biomedical informatics have sought to address this by exploring temporal point processes for EHR trajectory modeling and advanced computational phenotyping [1, 2].

Building upon this foundation, we propose casting the clinical challenge of early disease detection as a rigorous sequential recommendation (SR) problem. In recent years, SR has emerged as a new paradigm that takes historical behavior trajectories as input, aiming to predict the next behavior object an entity will interact with. We propose reconstructing patients and attributes at different time points into dynamic graph nodes, mapping raw patient-visit interactions (such as treatment counts) into graph edge weights. A time-sensitive graph edge prediction based on a sequential recommendation system serves as a powerful candidate on our restructured EHR data to detect the potential link between a given patient and a rare disease of interest based on historical graph structures.

Building a technically robust dynamic graph for this purpose requires overcoming several algorithmic limitations. Dynamic graphs leverage not only structural but also temporal patterns to enable more powerful graph representations. However, most existing models fail to model the temporal dynamics of patients and visits simultaneously, thus ignoring the high-order

¹Advanced Analytics, IQVIA Inc., Wayne, PA, USA.

*email: guanhao.wei@iqvia.com

collaborative information between them. Although recent works have tried to apply dynamic graph-based approaches to sequential recommendation [3–5], they are limited because graph neural networks (GNNs) are inherently localized and inclined to have over-smoothed representations [6]. One possible solution is to mine micrograph structures to characterize nodes by their local connectivity and subgraph/motif patterns [7–9]. We hypothesize that enhancing continuous-time embedding with motif patterns in the discrete-time domain will significantly improve model performance in sequential recommendation.

Furthermore, applying this system to public health data requires addressing the inherent nature of clinical records. Medical records usually only contain information about diseases a patient has been diagnosed with, and do not include information stating which diseases a patient does not have. Therefore, Positive-unlabeled (PU) learning and weak supervision techniques naturally fit this significant real-world scenario [10], operating on the assumption that each unlabeled example could belong to either the positive or negative class. Previous works [11–14] have shown that additional hard negative samples improve model performance. In this work, we focus on developing a rigorous PU-learning approach through graph-based negative sampling, effectively utilizing rich graph structures to identify reliable negative examples [15].

To address all of the challenges listed above, we propose a Dynamic graph-based PU-learning approach for Sequential Recommendation, termed DPU-SR. Our major contributions can be summarized as follows:

- We propose a novel dynamic graph learning framework for sequential recommendation, which models multi-granularity temporal patterns in an integrative context.
- We develop an efficient structure-aware negative sampling approach for dynamic graph learning.
- We demonstrate the effectiveness of the proposed algorithms on public datasets and deploy them on a large-scale, real-world rare disease dataset to detect multiple myeloma.

2. Materials and methods

2.1. Problem formulation

In standard epidemiological surveillance, patient data are often treated cross-sectionally. However, to capture the evolving clinical presentation of rare diseases, we formulate longitudinal patient trajectory forecasting as a continuous-time dynamic graph problem.

Within our sequential recommendation framework, we define the entire patient population as U and the universe of distinct clinical events (such as diagnoses, procedures, or prescriptions) as V . For any given individual patient u , their medical history is represented as a time-ordered sequence of events $S^{(u)} = (v_1, v_2, \dots, v_{n_u})$, which corresponds to an exact array of timestamps $T^{(u)} = (t_1, t_2, \dots, t_{n_u})$. The total number of events for this patient is given by n_u . Fundamentally, our objective is to determine the most probable subsequent clinical event (for instance, the onset

of a new disease) by analyzing these past interactions. Mathematically, this involves estimating the conditional probability $p(v_{n_u+1}|S^{(u)})$. The model then generates a ranked list of potential future visits sorted by this calculated likelihood.

This study frames the task of sequential clinical forecasting as a dynamic link prediction challenge. To achieve this, we map all historical patient–visit activities onto a continuous-time bipartite graph, denoted as $\mathcal{G}$, which comprises two distinct node categories: patients (U) and medical visits (V). Whenever patient u experiences clinical event v at a specific timestamp t , a bidirectional connection is established. This consists of a forward edge from u to v tracking the patient’s chronological event index o_u^v , and a reciprocal edge from v to u recording the patient’s position o_v^u among all individuals experiencing that specific event. By masking all interactions that occur after a given time t , we can extract a static graph snapshot $\mathcal{G}^t$. Consequently, forecasting the next event v_{n_u+1} for a specific sequence $S^{(u)}$ and timestamp array $T^{(u)}$ is mathematically equivalent to predicting the formation of a new edge connected to node u within the snapshot $\mathcal{G}^{t_{n_u}}$. To maintain computational efficiency, we truncate overly long sequences, considering only the k most recent clinical events $(v_{n_u-k+1}, \dots, v_{n_u})$ to make predictions when $n_u > k$.

Clinical walkthrough: To make this mathematical formulation accessible in a practical context, consider a patient’s longitudinal journey example, when a patient (node u) visits a clinic and is diagnosed with anemia (clinical event node v_1) on January 1st (timestamp t_1), a directed temporal edge (u, v_1, t_1) is established in the graph. If the patient is subsequently prescribed a treatment, such as Furosemide (clinical event node v_2), on February 15th (timestamp t_2), a new edge (u, v_2, t_2) is formed.

Rather than treating this medical history as a static, unordered list of billing codes, the dynamic graph constructs a chronological web of interactions. Crucially, the model does not just register that both events occurred; it uses the time sequence to explicitly calculate the exact 45-day interval between the symptom and the treatment. This enables the algorithm to capture the clinical velocity of disease progression, successfully translating a messy, real-world Electronic Health Record into structured mathematical inputs.

2.2. Time-aware clinical information propagation

Extracting predictive signals from vast EHR requires isolating relevant clinical histories. We apply a neighborhood sampling strategy to construct patient-specific dynamic subgraphs, capturing high-order sequence information from a patient’s recent medical events. Crucially, standard graph approaches often disregard exact temporal constraints. In clinical practice, however, the absolute timespan between two medical events—for instance, a rapid succession of emergency visits versus routine annual checkups—holds vital prognostic value for rare disease progression. Therefore, we employ a continuous-time embedding to explicitly parameterize the time intervals between clinical interactions. The proposed architecture leverages a bidirectional information propagation mechanism to capture how the temporal dynamics of interconnected patient sequences influence the risk assessment of a specific individual. **Figure 1** shows a general model architecture.

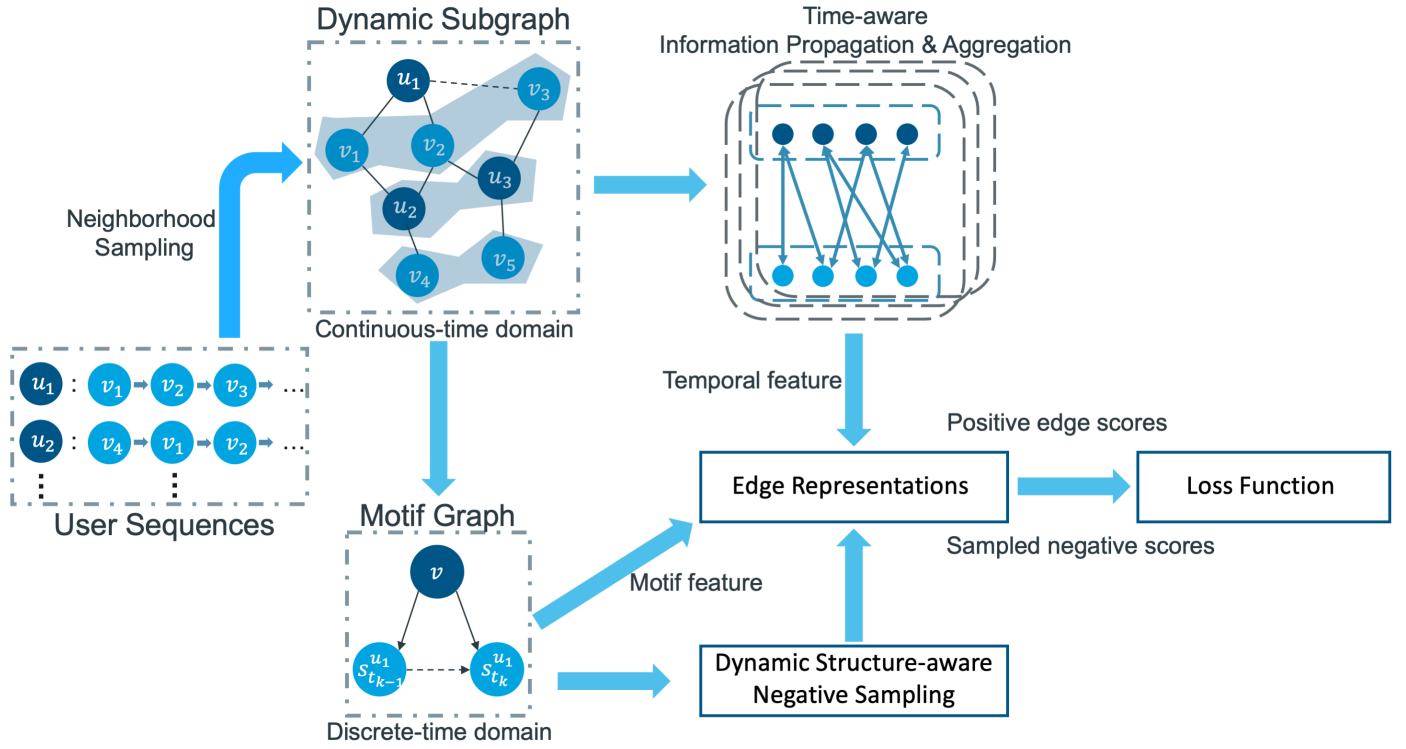


Figure 1 • Model architecture of our proposed Dynamic Graph-based Positive-Unlabeled Learning (DPU-SR) method. The framework maps sequential user patterns to structural embeddings through coordinated continuous and discrete-time components. Dark blue circles (u_1, u_2, u_3) represent patient nodes, light blue circles ($v_1, v_2, \dots, v_5$) represent item or medical event nodes, and the shaded blue regions enclose localized dynamic subgraphs. In the continuous-time domain, solid black lines indicate active dynamic interactions, while dashed lines signify historical or long-range connections. Within the time-aware propagation module, vertical directional arrows illustrate bipartite information exchange across temporal sequence layers. In the discrete-time domain (Motif Graph), variables $S_{t_{k-1}}^{u_1}$ and $S_{t_k}^{u_1}$ denote the structural states of a user at preceding and current discrete time steps t_{k-1} and t_k , interconnected by a horizontal dashed arrow representing temporal progression. Large light blue blocks indicate major computational phases, and thick light blue arrows trace the overall operational data flow leading to the generation of Edge Representations and final loss optimization.

2.2.1. Patient and visit embedding

Initial patient and visit node features are low-dimensional embedding matrices $H_U^0 \in \mathbb{R}^{|U| \times d}$ and $H_V^0 \in \mathbb{R}^{|V| \times d}$, where d is the embedding dimension. For each interaction between patient u and visit v , we retrieve the node embedding h_u^0 and h_v^0 by indexing embedding matrix. Each embedding matrix is initiated randomly and will be optimized during model training. To fully integrate non-sequential patient characteristics into the dynamic graph framework, the static demographic variables are directly concatenated to the initialized patient node embedding vector h_u^0 prior to the temporal information propagation steps.

2.2.2. Continuous-time embedding

Most of the graph representation learning approaches are designed for static graphs, and thus disregard temporal constraints and may mistakenly utilize time information. In order to better encode temporal effects in a dynamic graph, we plan to design a dynamic information propagation mechanism with continuous-time embedding. The continuous-time embedding behaves as a continuous functional mapping $\Phi: T \rightarrow \mathbb{R}^{d_T}$ from time domain $T \in \mathbb{R}^+$ to the d_T -dimensional vector space. Usually, the relative timespan, instead of the absolute value of time, reveals vital temporal effects in a sequential recommendation task [16]. Therefore, we apply the temporal kernel function to learn timespan patterns [17]. To be more precise, we characterize the temporal kernel $\mathcal{K}: T \times T \rightarrow \mathbb{R}$ using the following formulation

$$\mathcal{K}(t_1, t_2) = \Phi(t_1) \cdot \Phi(t_2) = \psi(t_1 - t_2). \quad (1)$$

Within this expression, the symbol $\cdot$ signifies the dot product, while the term $\psi(t_1 - t_2)$ is utilized to quantify the degree of temporal correlation between any two given timestamps. We can explicitly define the time embedding via the following formulation, a design choice grounded in Bochner's theorem [18] that ensures the kernel $\mathcal{K}$ remains continuous and translation-invariant:

$$\Phi(t) = \sqrt{\frac{1}{d_T}} [\cos(\omega_1 t), \sin(\omega_1 t), \dots, \cos(\omega_{d_T} t), \sin(\omega_{d_T} t)]. \quad (2)$$

where $\omega = [\omega_1, \dots, \omega_{d_T}]$ are learnable parameters.

2.2.3. Bidirectional information propagation

The proposed architecture performs bidirectional message passing directly within the extracted dynamic subgraphs, effectively linking patient entities with clinical visit nodes. At every network layer, the structural node vectors are augmented with temporal embeddings to construct the input features. To be exact, given a patient node u_i evaluated at a specific query timestamp t , the time-aware feature representation fed into the l -th layer is formulated as:

$$\tilde{h}_{u_i}^{(l-1)}(t) = h_{u_i}^{(l-1)}(t) \parallel \Phi(t). \quad (3)$$

In this equation, the operator $\parallel$ signifies vector concatenation, with the layer index denoted by $l \in \{1, 2, \dots, L\}$. For the initial

layer ($l = 1$), the term $h_{u_i}^{(l-1)}(t)$ corresponds to the raw, initialized node embedding, whereas for any subsequent layer ($l > 1$), it represents the hidden state derived from the preceding layer's output. During the propagation step at each layer, the temporal representations originating from the immediate neighborhood $\mathcal{N}_{u_i}^1(t)$ are aggregated into the target node u_i . Specifically, the message transmitted across a given interaction edge (u_i, v_j, t_k) is defined as:

$$\tilde{h}_{v_j}^{(l-1)}(t_k) = h_{v_j}^{(l-1)}(t_k) \parallel \Phi(t_k). \quad (4)$$

To effectively capture the intricate dependencies embedded within high-order structural connections, the framework utilizes a deep, multi-layered design. Through the iterative execution of this message-passing protocol, the target patient's final embedding successfully absorbs the temporal and structural context spanning its L -hop clinical neighborhood. From a conceptual standpoint, this mechanism functions to quantify how the chronological trajectories of structurally proximate patients influence the focal individual's evolving risk profile.

Unlike traditional recommendation systems that only focus on analyzing the impact of historical visits on patient preference, we propose to capture the effect of patient appearance on visit attributes. We generalize the above information propagation mechanism to visit nodes. Analogously, we construct time-aware visit representation and propagate patient neighborhood node information to its target visit nodes.

2.3. Clinical motif extraction

To model complex symptomology, our framework utilizes a temporal graph attention mechanism to weigh the clinical significance of historical events. Furthermore, to capture micro-structures in discrete time, we extract motif features. In the context of EHR data, graph motifs represent higher-order connectivity patterns, effectively acting as proxies for common clinical pathways or symptom clusters. By converting patient trajectories into a directed motif graph, we calculate the transition probabilities between consecutive medical events. This allows the model to recognize subtle, sequential patterns that frequently precede rare disease diagnoses.

2.3.1. Temporal graph attention mechanism

Given a patient node's time-aware node representation $\tilde{h}_{u_i}^{l-1}(t)$, the propagated information from its neighborhood nodes can be represented by a sequence:

$$Z_{u_i}^l(t) = [\tilde{h}_{v_1}^{(l-1)}(t_1), \tilde{h}_{v_2}^{(l-1)}(t_2), \dots, \tilde{h}_{v_N}^{(l-1)}(t_N)]. \quad (5)$$

where $\mathcal{N}_{u_i}^1(t) = (v_1, v_2, \dots, v_N)$. We can obtain the 'query', 'key', and 'value' with three different linear projections, which are the transformation matrices employed to depict the interactions between node embeddings and continuous-time embeddings. Subsequently, an attention mechanism based on dot-product operations is employed to compute a weighted aggregation across all respective entities within V :

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V. \quad (6)$$

Within this formulation, the relational interactions computed between the query and key pairs inherently dictate the distribution

of the attention weights. Beyond this linear attention step, it is crucial to model the complex, non-linear dependencies between the focal node's intrinsic attributes and the newly aggregated neighborhood states. To achieve this, we fuse these two sets of representations via vector concatenation. The combined output is then channeled through a dedicated aggregation module, structured as follows:

$$h_{u_i}^l = \text{ReLU}\left(\left[\text{Attn}(Q, K, V) \parallel \tilde{h}_{u_i}^{(l-1)}(t)\right] W_0^{(l)}\right). \quad (7)$$

Here, the trainable projection matrix is denoted by $W_0^l \in \mathbb{R}^{(d+d_T+d_h) \times d}$. The resulting vector $h_{u_i}^l$ represents the updated, refined embedding for the target patient node u_i at timestamp t upon completing the l -th layer's transformations. This entire attention and aggregation pipeline applies symmetrically when a clinical visit node serves as the query target. For such cases, one simply substitutes the focal patient's query vectors with the corresponding visit node's query data, adjusting the contextual neighborhood definitions detailed in **Equation (5)** to match the new target. Within our proposed network architecture, the embeddings for both the patient cohort and the clinical visit events are advanced and optimized concurrently during every single layer iteration.

2.3.2. Temporal feature

For each candidate edge (u_i, v_j, t_k) , we can obtain temporal node embeddings of both patient and visit nodes from the last model layer, denoted as $h_{u_i}^L(t_k)$ and $h_{v_j}^L(t_k)$ respectively. We define the temporal edge feature as follows:

$$e_{\text{temporal}}(u_i, v_j, t_k) = h_{u_i}^L(t_k) \cdot h_{v_j}^L(t_k). \quad (8)$$

The temporal edge feature characterizes the impacts of temporal collaborative signals among patient and visit nodes, which are tightly connected through multi-layer information propagation and aggregation.

2.3.3. Motif feature

Graph motifs are higher-order connectivity patterns of small subgraphs, which are crucial for understanding the complex graph structure. A motif has several topologically equivalent appearances in the network called motif instances. To capture the structure information among visit nodes in our dynamic graph, we convert patient sequences into a directed and weighted motif graph $\mathcal{G}_{\text{motif}}$. Each graph node $v \in \mathcal{V}_{\text{motif}}$ represents a visit and each edge denotes a directed transition between two nodes. We note that $\mathcal{V}_{\text{motif}}$ is a subset of $\mathcal{V}$ to reduce computational cost. The edge weight is calculated by transition probabilities defined as follows:

$$P(v_i, v_j) = \frac{\sum_{u \in \mathcal{U}} f(s_{t_{k-1}}^u = v_i, s_{t_k}^u = v_j)}{\sum_{u \in \mathcal{U}} f(s_{t_{k-1}}^u = v_i)}. \quad (9)$$

In this formulation, the variable $s_{t_k}^u$ corresponds to the specific clinical event that patient u experiences at timestamp t_k within their longitudinal history S^u . The function $f(s_{t_{k-1}}^u = v_i, s_{t_k}^u = v_j)$ quantifies the historical occurrence count of a sequential transition from event v_i directly to event v_j across the entire graph. This strict sequential ordering implies that the chronological index satisfies $o_u^{v_j} = o_u^{v_i} + 1$. Naturally, if no such transition pathway has ever been recorded between two nodes, the corresponding edge weight defaults to zero.

Our analysis specifically investigates three-node structural configurations, commonly referred to as open triad motifs, which can be categorized into the nine distinct topological classes illustrated in **Figure 2**. Because the core objective of our sequential recommendation task is to estimate the likelihood of a future connection emerging from a patient's preceding visit s_t^u to their subsequent encounter s_{t+1}^u , the analytical focus is placed heavily on triad structures that encompass these two pivotal nodes. Consequently, for any specific motif type n , we define the corresponding set of informative instances as:

$$M_n(s_{t_{k-1}}^u, s_{t_k}^u) = \left[(s_{t_{k-1}}^u, v, s_{t_k}^u) \right]. \quad (10)$$

where v is the intermediate node, which can be any visit in $\mathcal{G}_{motif}$. Each motif instance can be represented by a four-dimensional vector $h_{motif}(s_{t_{k-1}}^u, v, s_{t_k}^u)$ as follows:

$$[P(s_{t_{k-1}}^u, v), P(v, s_{t_k}^u), P(s_{t_k}^u, v), P(v, s_{t_{k-1}}^u)]. \quad (11)$$

Then, we obtain the representation $\mathcal{H}_{(s_t^u, s_{t+1}^u)}^k \in \mathbb{R}^4$ of each type- n motif:

$$\mathcal{H}_{(s_{t_{k-1}}^u, s_{t_k}^u)}^k = \sum_{v \in \mathcal{V}} h_{motif}(s_{t_{k-1}}^u, v, s_{t_k}^u). \quad (12)$$

We generate the motif features of edge (u, s_{t+1}^u) in $\mathcal{G}$ through concatenating all types of motif representations:

$$e_{motif}(u, s_{t_k}^u) = \mathcal{H}_{(s_{t_{k-1}}^u, s_{t_k}^u)}^1 \parallel \cdots \parallel \mathcal{H}_{(s_{t_{k-1}}^u, s_{t_k}^u)}^9. \quad (13)$$

Finally, for edge (u_i, v_j, t_k) , we aggregate the temporal feature and motif feature and output a score scalar through a two-layer fully connected network (FCN):

$$s(u_i, v_j, t_k) = \text{FCN}(e_{temporal}(u_i, v_j, t_k) \parallel e_{motif}(u_i, v_j)). \quad (14)$$

The score captures how likely it is that u_i will interact with v_j at time t_k . It is important to note the generalizability of this motif framework. While the mathematical structure of these open triad motifs is highly generalizable and can be extracted from any longitudinal EHR sequence, the specific transition probabilities and motif feature weights learned by the FCN are inherently disease-specific. Applying this framework to a different rare disease would require retraining the motif graph to capture the unique clinical pathways and symptom clusters associated with that specific condition.

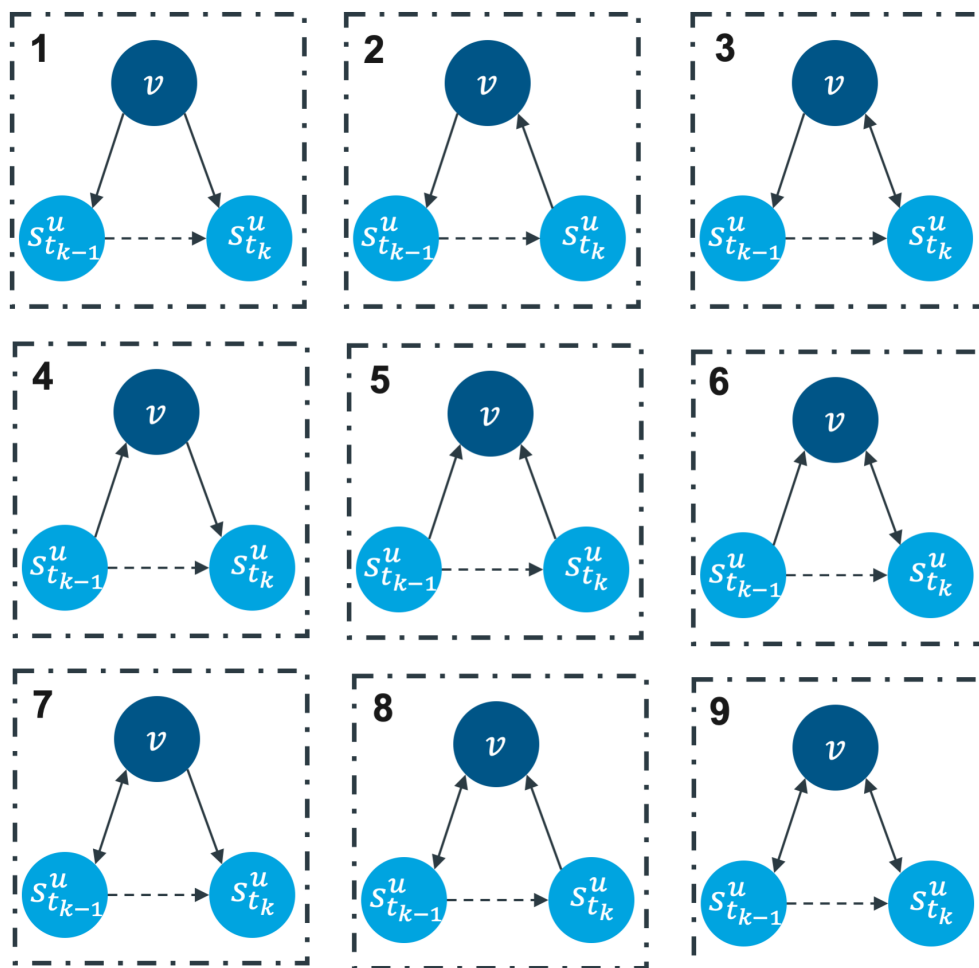


Figure 2 • Nine types of open triad motifs. The dark blue circle labeled v represents an item or clinical disease state node, while the light blue circles labeled $s_{t_{k-1}}^u$ and $s_{t_k}^u$ denote the patient's sequential event nodes at discrete time steps t_{k-1} and t_k , respectively. Solid arrows indicate directed active relations or transitions between the item node and the patient events, whereas the horizontal dashed arrow signifies the forward temporal progression of the patient from step t_{k-1} to t_k . The labels 1 to 9 categorize the nine unique topological configurations of open triad motifs formed by varying the directions of the solid edges connecting node v to the patient sequence, enumerating all possible interaction patterns across the discrete-time domain.

2.4. Addressing imperfect EHR data via PU-learning

A primary challenge in utilizing EHR data for public health is its Positive-Unlabeled (PU) nature; records confirm diagnoses that a patient has received, but rarely document the definitive absence of a condition. Treating all unrecorded diseases as strictly negative severely biases early detection models. To address this, we introduce dynamic structure-aware negative sampling. The model identifies reliable negative examples by locating clinical event nodes that exist within a patient's broader neighborhood but share no direct temporal relationship or clinical motif similarity with their actual medical journey. This allows the algorithm to learn robust decision boundaries without falsely assuming unlabeled patients are entirely disease-free.

Initially, we assume that unconnected visit nodes within the target user's broader neighborhood exhibit strong inter-connectivity, which makes them ideal candidates to be sampled as robust negative examples. Following this logic, the candidate pool for negative examples, denoted as $\mathcal{V}_{sample}$, is strictly constrained to the k -hop structural neighborhood of the focal patient node. Specifically, after constructing dynamic subgraph $\mathcal{G}(t)$ for a target patient u , we randomly sample visit nodes from their third-order neighborhood nodes $\mathcal{N}_u^3(t)$. We note that $\mathcal{N}_u^1(t)$ are visit nodes that share direct interactions with patients, which need to be excluded based on our hypothesis, and $\mathcal{N}_u^2(t)$ are related patient nodes based on our neighborhood sampling strategy. By selecting from this third-order neighborhood, we deliberately sample clinical events that are structurally connected within the broader graph (i.e., medical visits belonging to similar patients) but reside strictly outside the target patient's direct trajectory. This strategy provides robust "hard negatives" that prevent the model from learning trivial decision boundaries.

To enrich the negative sampling process with temporal information and find harder negative examples for effective graph learning, we make further selections from the visit subset obtained from the first step. Specifically, For a positive edge (u, s_{t+1}^u) , we construct the motif graph with $\mathcal{V}_{motif} = \mathcal{V}_{sample} \cup \{s_t^u, s_{t+1}^u\}$. We calculate motif features of each negative candidate edge (u, v) where $v \in \mathcal{V}_{sample}$, and compute the corresponding cosine similarity with the positive edge motif feature. We finally choose negative visits $\mathcal{V}_{neg}$ with top- k similarity values; we set $k = 3$ as the value in our experiments.

3. Results

3.1. Algorithmic validation on standard benchmarks

Given an observed positive edge $(u_i, s_{t_k}^{u_i})$ with negative edge samples (u_i, v_j, t_k) where $v_j \in \mathcal{V}_{neg}$, we train the model using the following loss function in an end-to-end fashion:

$$\begin{aligned} \mathcal{L} = & -\log \sigma \left(\gamma - s(u_i, s_{t_k}^{u_i}, t_k) \right) \\ & - \sum_{j=1}^{|\mathcal{V}_{neg}|} \frac{1}{|\mathcal{V}_{neg}|} \log \sigma (s(u_i, v_j, t_k) - \gamma). \end{aligned} \quad (15)$$

Prior to deployment on restricted clinical data, we validated the mathematical robustness of DPU-SR using four standard, highly sparse public benchmark datasets (MovieLens-1M [19] and Amazon CDs, Games, and Beauty [20]). These datasets serve as a proxy to verify the model's capacity to process multi-granularity temporal patterns and extract micro-structures from dynamic sequences.

To adequately model the temporal graph dynamics, every chronological patient trajectory is partitioned into a collection of sequential subsets paired with their respective target labels. Consider a hypothetical trajectory $\mathcal{S}^{(u)} = (v_1, v_2, v_3, v_4, v_5)$; this is divided into rolling windows where the immediate subsequent clinical event serves as the predictive target. Consequently, the network constructs corresponding graph snapshots alongside their future node connections, such as predicting v_4 based on snapshot $\mathcal{G}^{t_3}$, or predicting v_5 given $\mathcal{G}^{t_4}$.

The predictive accuracy of the framework is quantified utilizing two standard ranking indicators: Hit@10 and Normalized Discounted Cumulative Gain at 10 (NDCG@10). Specifically, Hit@10 measures the proportion of actual true clinical events that successfully appear within the model's top 10 prioritized predictions. Meanwhile, NDCG@10 provides a position-aware assessment, effectively rewarding accurate predictions that appear higher in the ranked output list by applying a logarithmic discount to those that are placed lower.

The proposed DPU-SR architecture was benchmarked against a diverse array of established sequential recommendation algorithms, including Bayesian Personalized Ranking based Matrix Factorization (BPR-MF) [21], Factorized Personalized Markov Chains (FPMC) [22], the Enhanced Gated Recurrent Unit for Recommender Systems (GRU4Rec+) [23], Convolutional Sequence Embedding Recommendation (Caser) [24], Time Interval Aware Sequential Recommendation (TiSASRec) [25], Self-Attention based Sequential Recommendation (SASRec) [26], and Hierarchical Gating Networks (HGN) [27].

Across all publicly available evaluation sets, our framework demonstrated superior predictive capability. Notably, DPU-SR yielded the most robust outcomes across both ranking criteria, eclipsing the most competitive baseline by an absolute margin of up to 3.77% in the NDCG@10 evaluation (comprehensive performance details are provided in **Table A1**).

3.2. Public health application: early detection of multiple myeloma

To validate the public health utility of our proposed DPU-SR framework, we transitioned from algorithmic benchmark testing to a large-scale epidemiological application: the early detection of newly diagnosed multiple myeloma (NDMM). Multiple myeloma is a type of rare tumor where delayed diagnosis severely impacts patient survival and limits treatment options. The modeling goal was to predict whether a candidate of interest would be diagnosed with multiple myeloma in the near future based on their historical medical journey. We deployed our algorithm on a large-scale, private real-world dataset comprising longitudinal medical service records for patients from IQVIA Anonymized Patient-Level Data (APLD), which is a high-granularity, longitudinal dataset tracking de-identified patient journeys across pharmacy and medical claims.

3.2.1. Clinical covariates and dynamic graph construction

To construct the dynamic temporal graph, we converted patients' medical services data into nodes and edges. The edges represented the time-stamped interactions between patients and specific medical events, primarily encompassing prescriptions (rx), medical procedures (px), and recorded diagnoses (dx). The

model ingested a rich set of clinical covariates, specifically utilizing 159 aggregated dx, rx, and px features, alongside 30 demographic variables for each patient. **Figure 3** visualizes a patient's historical medical journey that we captured before the first MM diagnosis. Crucially for forecasting disease progression, these patient attributes were updated monthly with clear time-stamps attached, allowing the dynamic graph to capture the exact chronological ordering and time intervals of a patient's evolving symptomatology.

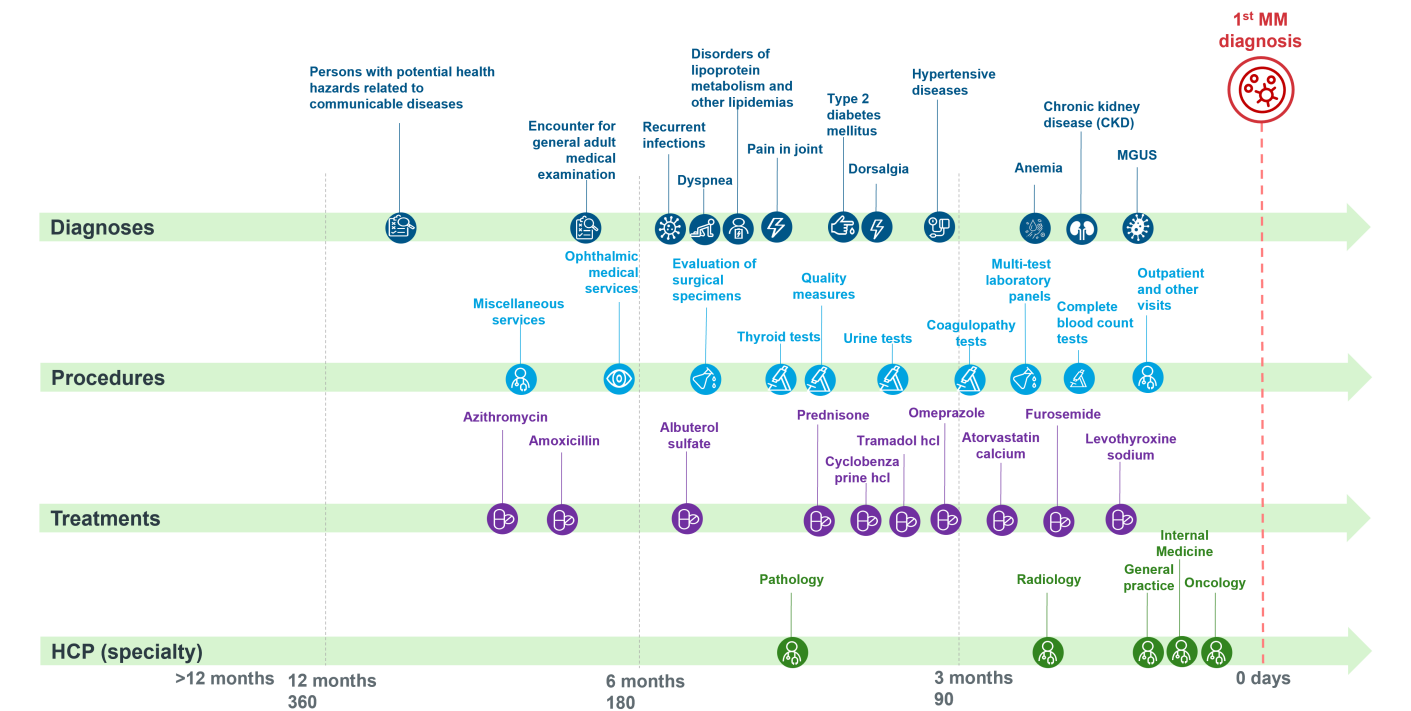


Figure 3 • Sample sequential Electronic Health Record (EHR) records of a new multiple myeloma (MM) diagnosis patient. The horizontal timeline tracks the longitudinal patient journey across four distinct clinical streams differentiated by color: dark blue circles indicate diagnoses, bright blue circles denote clinical procedures, purple circles represent medication treatments, and dark green circles illustrate the healthcare professional (HCP) specialty involved. Standard medical clinical abbreviations embedded within the timeline are defined as follows: CKD stands for Chronic Kidney Disease, MGUS stands for Monoclonal Gammopathy of Undetermined Significance, and MM denotes Multiple Myeloma. The prominent vertical red dashed line on the far right explicitly marks the baseline timestamp (0 days) of the patient's first confirmed multiple myeloma (1st MM) diagnosis, serving as the target event for detection forecasting.

3.2.2. Experimental setup and evaluation protocol

To evaluate the model's prospective utility, we started by utilizing a massive longitudinal cohort comprising approximately 1.5 million unique patients who recorded doctor's visits between December 2017 and September 2020. To ensure clinical relevance, this screening population was specifically filtered to include individuals exhibiting early-stage symptoms akin to the typical multiple myeloma (MM) patient journey. Ultimately, only about 0.6% of these candidates were definitively diagnosed with MM, underscoring the massive class imbalance inherent to rare disease surveillance.

For the purposes of predictive modeling, a positive patient was strictly defined as an individual receiving their first confirmed MM diagnosis within the immediate next quarter (a three-month window following the index date). This definition yielded approximately 250 to 300 incident positive cases per month. And given the index date, the total number of unlabeled patients with at least 10 medical events (diagnosis/procedure/drug use) in the 2-year

lookback period will reach up to around 105,000 per month. So the average positive ratio in a monthly cohort is 0.28%. Our model was trained using four carefully selected monthly data snapshots: March 2018, June 2018, September 2018, and March 2019. Because each training snapshot is separated by a gap of at least three months, the defined positive patients are entirely unique across the four cohorts. While patients with an unknown diagnostic status may remain in the dataset across multiple training snapshots, their extracted clinical features dynamically evolve to reflect the specific, updated temporal context in which they were captured.

To rigorously validate the framework's real-world generalizability and strictly prevent temporal label leakage, our testing cohorts were drawn exclusively from the 2020 calendar year. The significant temporal lag between the final training cohort (March 2019) and the testing period ensures that future clinical data from positive training cases cannot inadvertently influence the testing evaluation.

We deliberately selected Precision@Top 1000 as our primary operational metric. Under this protocol, the entire unlabeled testing cohort is ranked by the model's generated risk score, and precision is calculated strictly within the top 1000 highest-risk individuals. This metric mirrors a realistic, constrained clinical review budget where a healthcare system only has the resources to manually review a fixed number of alerts.

To ensure a rigorous and fair evaluation, all tested models—including classical machine learning baselines (Logistic Regression, Random Forest, XGBoost), sequential models (LSTMs), and our proposed DPU-SR framework—were anchored to the exact same raw medical events, feature universe (159 clinical covariates and 30 demographic variables), and temporal observation windows. However, because these algorithms possess fundamentally different structural architectures, the raw medical events required distinct preprocessing transformations. For example, while XGBoost ingested flattened, aggregated frequency counts of these events over the observation window, the LSTM processed them as ordered chronological vectors, and DPU-SR transformed the exact same underlying events into dynamic temporal subgraphs. This

strict control ensures that any observed performance lift is strictly attributable to the model's architectural capacity to capture complex clinical trajectories, rather than discrepancies in underlying data access.

3.2.3. Performance and impact

Our proposed dynamic graph-based recommendation system successfully predicted the existence of future links between a patient node and an MM diagnosis node. Among the top 1000 scored high-risk patients, the DPU-SR model achieved an average precision of 6.3%. When contrasted against the classical production models currently utilized by the healthcare industry, like XGBoost or LSTM, our proposed approach demonstrated a 20% performance improvement (see **Figure 4**).

This massive boost in precision illustrates a significant advancement for public health practice. By effectively finding the “needle in the haystack” within messy, complex EHR data, this AI framework provides health systems with an actionable tool to proactively identify high-risk individuals, reduce diagnostic delays, and optimize the allocation of specialized hematological resources.

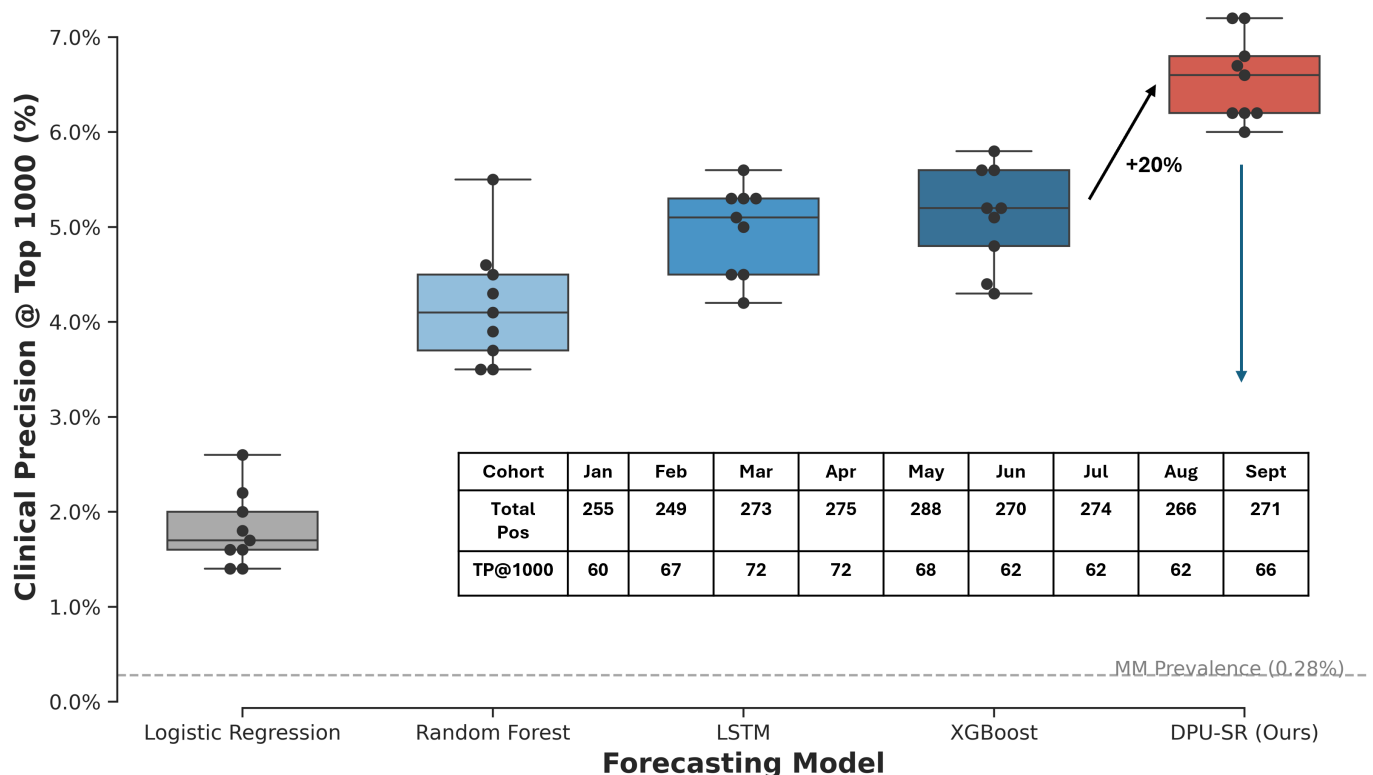


Figure 4 • Dynamic PU-Learning Approach for Sequential Recommendation (DPU-SR) model performance compared with classical production models in the industry. The boxplot tracking clinical precision at the top 1,000 predicted high-risk individuals (Clinical Precision @ Top 1000 (%)) illustrates model performance, where each individual black dot represents an independent evaluation cohort score across different monthly observation windows (from January to September). The horizontal lines within the boxes indicate the median precision, the box boundaries represent the interquartile range (IQR), and the error bars (whiskers) extend to the minimum and maximum data points. Color codes differentiate the model types: gray for Logistic Regression, light blue for Random Forest, medium blue for Long Short-Term Memory (LSTM) and Extreme Gradient Boosting (XGBoost), and coral red for our proposed DPU-SR model, which exhibits a 20% performance improvement over the best baseline. The baseline gray dashed line denotes the baseline Multiple Myeloma (MM) prevalence (0.28%). The embedded table details the monthly patient counts, where “Total Pos” represents the total number of true positive patients in that month’s cohort, and “TP@1000” represents the absolute number of True Positives correctly identified within the top 1,000 scored patients; all values in the table are absolute patient counts.

4. Discussion

4.1. Transforming clinical trajectories into actionable surveillance

The early detection of rare diseases represents a profound public health challenge, as these conditions affect over 300 million individuals globally but suffer from chronically high misdiagnosis rates due to their complex and varied clinical presentations. In this study, we reframed the clinical challenge of early disease detection as a dynamic sequence recommendation problem, demonstrating that advanced graph neural networks can effectively map longitudinal patient trajectories. By conceptualizing patients as nodes and their historical medical events (including diagnoses, procedures, and prescriptions) as dynamic edges, our DPU-SR framework successfully captures the temporal evolution of a patient's medical journey. This approach moves beyond static risk-scoring models, providing public health practitioners with a mechanism to identify high-risk patients based on the nuanced, multi-granularity timing of their clinical interactions [28, 29].

4.2. Addressing the reality of imperfect public health data

A significant barrier to deploying artificial intelligence in public health practice is the inherent nature of real-world Electronic Health Records. Medical records inherently catalog the presence of a diagnosis but rarely confirm the definitive absence of one. Standard binary classification models struggle in this environment. Our implementation of a Positive-Unlabeled (PU) learning approach directly addresses this data asymmetry, aligning with recent methodologies designed to tackle the lack of reliable control groups and partially observed outcomes in observational databases [30, 31]. By utilizing dynamic structure-aware negative sampling, the DPU-SR model identifies reliable negative examples without falsely assuming that unlabelled patients are entirely disease-free. This methodological innovation is highly applicable to broader epidemiological surveillance, where public health databases frequently suffer from incomplete or right-censored patient timelines. From an operational perspective, effectively managing this Positive-Unlabeled data enables health systems to proactively surface undiagnosed patients who are currently lost in these clinical data gaps. Rather than waiting for a definitive, late-stage diagnostic code, practitioners can use these dynamic risk profiles to flag patients for earlier, targeted screening, thereby bridging the gap between passive record-keeping and proactive clinical intervention.

4.3. Real-world impact on advanced healthcare analytics

The transition from theoretical benchmarking to real-world application validates the public health utility of this framework. When deployed on a large-scale, private dataset of candidates to predict newly diagnosed multiple myeloma, our model greatly improved the Precision@Top 1000 performances compared to the widely accepted industrial production algorithms. For public health infrastructure and healthcare systems, integrating this kind of dynamic graph-based surveillance into population health management pipelines offers substantial predictive enrichment and the potential to inform earlier interventions and optimized allocation of specialized medical resources.

4.4. Limitations and future directions

While the DPU-SR model demonstrates significant predictive power, its reliance on dense, longitudinal EHR data means its efficacy may be reduced in under-resourced health systems where patient records are fragmented. Future work should explore the integration of multimodal public health data, such as social determinants of health (SDoH) or genomic profiles, into the dynamic subgraph to further refine the motif features and improve early detection accuracy across diverse patient populations [32, 33].

Crucially, because this is a retrospective analysis, our findings demonstrate predictive enrichment rather than causal clinical outcomes. Prospective clinical trials integrated into live diagnostic workflows are ultimately required to definitively prove the existence of actual improvements in diagnosis timing, patient survival, and systemic resource allocation.

5. Conclusions

In this study, we presented a novel Dynamic Graph-Based Positive-Unlabeled Learning approach to forecast longitudinal patient trajectories and improve the early detection of rare diseases. Our framework, DPU-SR, effectively models the complex, multi-granularity temporal patterns of a patient's medical journey by transforming clinical events into dynamic subgraphs. Furthermore, by implementing a dynamic structure-aware negative sampling technique within a PU learning setting, our model successfully navigates the inherent right-censored nature of real-world EHR, learning from confirmed diagnoses without falsely assuming unlabeled patients are entirely disease-free. Ultimately, translating sequential EHR data into dynamic graphs provides health systems with a powerful, scalable tool for epidemiological surveillance. By identifying high-risk patients earlier and more accurately, this framework demonstrates strong predictive enrichment and the potential to inform early intervention. While prospective clinical validation is necessary to confirm long-term outcomes, this AI-driven approach provides health systems with a scalable foundation for proactive epidemiological surveillance, directly supporting the core goals of modern public health practice.

Acknowledgments

The authors thank Zichen Wang for his foundational efforts during the initial stages of this research when he was working at the University of California, Los Angeles (UCLA). His contributions to the algorithmic design and computational execution were integral to the development of the framework presented in this paper.

Funding

This research received no external funding.

Author contributions

Conceptualization, G.W.; methodology, G.W.; validation, G.W. and Y.W.; resources, L.Z.; data curation, G.W.; writing—original draft preparation, G.W.; writing—review and editing, G.W., Y.W. and L.Z.; project administration, Y.W. All authors have read and agreed to the published version of the manuscript.

Conflict of interest

Authors Guanhao Wei, Yunlong Wang, and Li Zhou are employed at IQVIA Inc. IQVIA Inc. provided the Anonymized Patient-Level Data (APLD) for this research, but played no role in the study's conception, design, execution, or interpretation, or with the writing of this manuscript. The authors declare no other financial or personal competing interests.

Data availability statement

The data supporting the findings of this publication cannot be made available due to strict privacy, legal, and proprietary restrictions surrounding the patient medical records provided by IQVIA Inc. For algorithmic validation, the publicly available benchmark datasets used in this study are openly accessible online: the MovieLens dataset is available via GroupLens Research at <https://grouplens.org/datasets/movielens/latest/>, and the Amazon review datasets (Beauty, Games, and CDs) are available at <https://nijianmo.github.io/amazon/#subsets>.

Institutional review board statement

Ethical review and approval were waived for this study, due to the use of a pre-existing, strictly de-identified longitudinal commercial database (IQVIA Anonymized Patient-Level Data) that fully complies with HIPAA privacy regulations, thereby not meeting the definition of human subjects research.

Appendix A

Table A1 • Model performance on standard benchmarks.

Dataset	Metric	BPR-MF	FPMC	GRU4Rec+	Caser	TiSASRec	SASRec	HGN	DPU-SR
MovieLens-1m	NDCG@10	0.3421	0.3917	0.4334	0.5011	0.5706	<u>0.5905</u>	0.5457	0.6002
MovieLens-1m	Hit@10	0.5952	0.6182	0.6522	0.7517	0.8038	<u>0.8245</u>	0.7663	0.8309
Amazon-beauty	NDCG@10	0.2183	0.2475	0.2552	0.2547	0.2818	0.3219	0.3241	<u>0.3221</u>
Amazon-Beauty	Hit@10	0.3775	0.3762	0.4048	0.4264	0.4345	<u>0.4854</u>	0.4853	0.4856
Amazon-Games	NDCG@10	0.2884	0.4481	0.4685	0.4694	0.4797	<u>0.5360</u>	0.4915	0.5562
Amazon-Games	Hit@10	0.4734	0.6452	0.6619	0.6865	0.7083	<u>0.7410</u>	0.7104	0.7587
Amazon-CDs	NDCG@10	0.3546	0.3285	0.4556	0.4689	<u>0.5047</u>	0.5023	0.4954	0.5224
Amazon-CDs	Hit@10	0.5534	0.5142	0.6758	0.6867	<u>0.7212</u>	0.7187	0.7152	0.7290

The best performing method in each row is boldfaced, and the second best method in each row is underlined. Metrics: NDCG@10 stands for Normalized Discounted Cumulative Gain at rank 10, and Hit@10 denotes Hit Ratio at rank 10. Baselines: BPR-MF represents Bayesian Personalized Ranking based Matrix Factorization; FPMC denotes Factorized Personalized Markov Chains; GRU4Rec+ is the Enhanced Gated Recurrent Unit for Recommender Systems; Caser represents Convolutional Sequence Embedding Recommendation; TiSASRec stands for Time Interval Aware Sequential Recommendation; SASRec denotes Self-Attention based Sequential Recommendation; HGN represents Hierarchical Gating Networks; and DPU-SR is our proposed Dynamic Graph-based PU-Learning Approach for Sequential Recommendation.

Additional information

Received: 4 March 2026

Accepted: 16 May 2026

Published: 2 June 2026

Academia AI and Applications papers should be cited as *Academia AI and Applications 2026*, ISSN 3071-0286, <https://doi.org/10.20935/AcadAI8326>. The journal's official abbreviation is *Acad. AI*.

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